

9.4.2.2.1 Mahalanobis distance-based separability measure

$$\begin{array}{lll} Q: & C_1 & (1,2) \quad (2,3) \quad (2,1) \\ & C_2 & (4,2) \quad (3,4) \quad (4,3) \end{array}$$

Calculate $J_{1,2} = ?$

$$\begin{aligned} \text{Solution } \textcircled{1} m_1 &= \left(\frac{1+2+2}{3}, \frac{2+3+1}{3} \right) \\ &= \left(\frac{5}{3}, 2 \right) \end{aligned}$$

$$\begin{aligned} m_2 &= \left(\frac{4+3+4}{3}, \frac{2+4+3}{3} \right) \\ &= \left(\frac{11}{3}, 3 \right) \end{aligned}$$

$$\begin{aligned} m &= \left(\frac{1+2+2+4+3+4}{6}, \frac{2+3+1+2+4+3}{6} \right) \\ &= \left(\frac{8}{3}, \frac{5}{2} \right) \end{aligned}$$

$$m_1 - m_2 = \left(\frac{5}{3}, 2 \right) - \left(\frac{11}{3}, 3 \right) = (-2, -1)$$

$$\begin{aligned}
(2) \quad C &= \frac{1}{6} \sum_{k=1}^6 (X_k - m)(X_k - m)^T \\
&= \frac{1}{6} \left[\begin{pmatrix} -\frac{5}{3} \\ -\frac{1}{2} \end{pmatrix} \begin{pmatrix} -\frac{5}{3} & -\frac{1}{2} \end{pmatrix} + \begin{pmatrix} -\frac{2}{3} \\ \frac{1}{2} \end{pmatrix} \begin{pmatrix} -\frac{2}{3} & \frac{1}{2} \end{pmatrix} + \begin{pmatrix} -\frac{2}{3} \\ -\frac{1}{2} \end{pmatrix} \begin{pmatrix} -\frac{2}{3} & -\frac{1}{2} \end{pmatrix} \right. \\
&\quad \left. + \begin{pmatrix} \frac{4}{3} \\ -\frac{1}{2} \end{pmatrix} \begin{pmatrix} \frac{4}{3} & -\frac{1}{2} \end{pmatrix} + \begin{pmatrix} \frac{1}{3} \\ \frac{1}{2} \end{pmatrix} \begin{pmatrix} \frac{1}{3} & \frac{1}{2} \end{pmatrix} + \begin{pmatrix} \frac{4}{3} \\ \frac{1}{2} \end{pmatrix} \begin{pmatrix} \frac{4}{3} & \frac{1}{2} \end{pmatrix} \right] \\
&= \frac{1}{6} \left[\begin{bmatrix} \frac{25}{9} & \frac{5}{6} \\ \frac{5}{6} & \frac{1}{4} \end{bmatrix} + \begin{bmatrix} \frac{4}{9} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{1}{4} \end{bmatrix} + \begin{bmatrix} \frac{4}{9} & 1 \\ 1 & \frac{9}{4} \end{bmatrix} \right. \\
&\quad \left. + \begin{bmatrix} \frac{16}{9} & -\frac{2}{3} \\ -\frac{2}{3} & \frac{1}{4} \end{bmatrix} + \begin{bmatrix} \frac{1}{9} & \frac{1}{2} \\ \frac{1}{2} & \frac{9}{4} \end{bmatrix} + \begin{bmatrix} \frac{16}{9} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{4} \end{bmatrix} \right] \\
&= \frac{1}{6} \begin{bmatrix} \frac{22}{3} & 2 \\ 2 & \frac{11}{2} \end{bmatrix} \\
&= \begin{bmatrix} \frac{11}{9} & \frac{1}{3} \\ \frac{1}{3} & \frac{11}{12} \end{bmatrix} \\
C^{-1} &= \begin{bmatrix} \frac{99}{109} & -\frac{36}{109} \\ -\frac{36}{109} & \frac{132}{109} \end{bmatrix}
\end{aligned}$$

$$J_{12} = (m_1 - m_2)^T C^{-1} (m_1 - m_2)$$

$$= (-2, -1) \begin{bmatrix} \frac{99}{109} & -\frac{36}{109} \\ -\frac{36}{109} & \frac{132}{109} \end{bmatrix} \begin{pmatrix} -2 \\ -1 \end{pmatrix}$$

$$= 3.5229.$$